

# STRUTHIO ONE

Build manual · Windows 11 · every step in build order



From ordering the parts to playing: flash the full game, build it, test it.  
Design locked · commit ba90005 · 2026-10-02

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## How to read this manual

Do the parts in order. Each step has a picture, what to do, and a green ✓ **check**: don't go on until it's true. Orange boxes are the things that matter most.

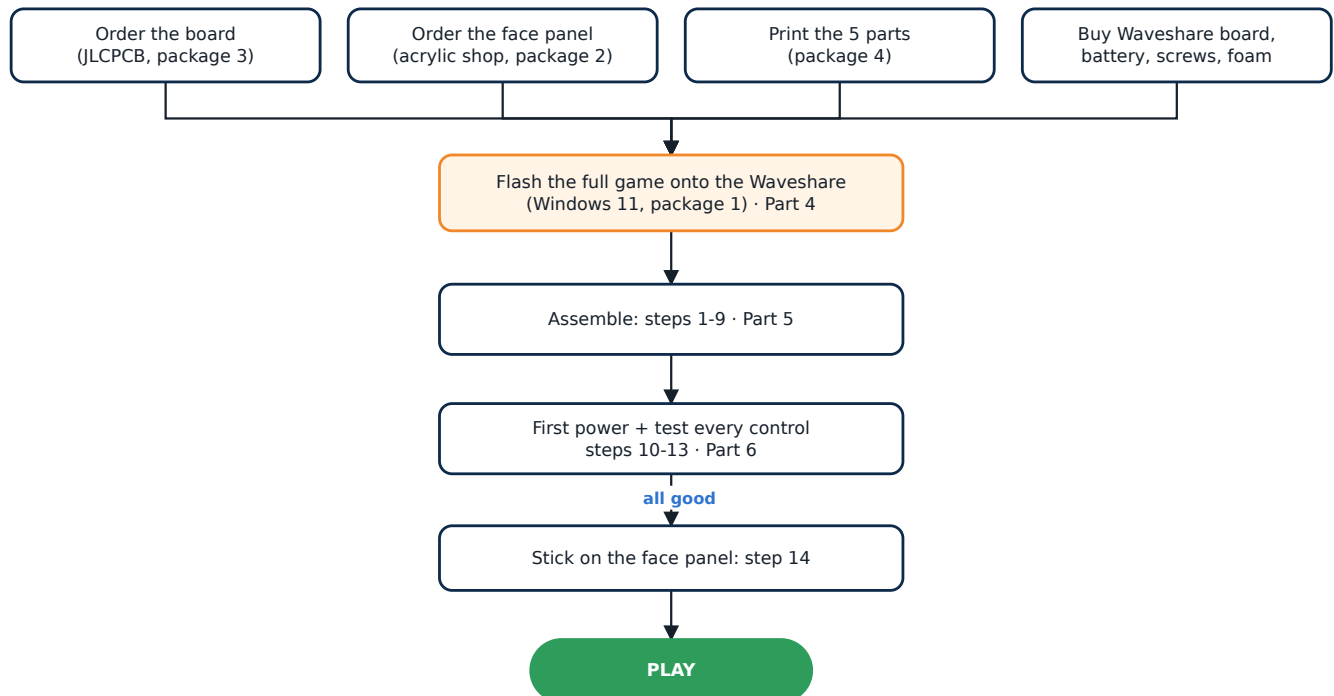
Pictures are labelled: **SCREENSHOT** real output of the real firmware and tools, **FROM THE 3D MODEL** rendered from the exact case and board design (the parts each step adds glow orange), **DIAGRAM** drawn from the design's own numbers, **ILLUSTRATION** a drawing of a Windows window to show where to click.

The six packages you need are referred to by number: **1** firmware flashing, **2** face panel print, **3** PCB (JLCPCB), **4** 3D print, **5** documents (this manual), plus the full backup.

# The whole build at a glance

## PART 1

Four things are made or bought at the same time; you flash the game as soon as the Waveshare board arrives, then build, test, and finish with the face panel.



The build, start to finish DIAGRAM



Everything that goes into it, front to back: face panel, front shell, buttons, Waveshare board, battery, ONE board, back shell FROM THE 3D MODEL

# What you need

## PART 2

### Parts

| ✓ Item                                                            | Exactly what                                                                                                         |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> <b>Waveshare ESP32-S3-Touch-LCD-3.5B</b> | The bare board (not the "-C" cased version). Its 6 Ω 1 W speaker comes with it.                                      |
| <input type="checkbox"/> <b>LiPo battery</b>                      | THOR-503450 (5 × 34 × 52 mm), 1000 mAh, ordered with a <b>JST PH 2.0 mm</b> 2-pin plug and an <b>80-100 mm</b> lead. |
| <input type="checkbox"/> <b>ONE board</b>                         | From JLCPCB, assembled (Part 3).                                                                                     |
| <input type="checkbox"/> <b>Face panel</b>                        | Laser-cut, back-printed 1.0 mm acrylic (Part 3).                                                                     |
| <input type="checkbox"/> <b>5 printed parts</b>                   | Front shell, back shell, two wing buttons, rocker (Part 3).                                                          |
| <input type="checkbox"/> <b>2 screws</b>                          | M2 × 8 pan head, self-tapping or machine.                                                                            |
| <input type="checkbox"/> <b>1.75 mm filament</b>                  | One 11 mm piece: the rocker's axle.                                                                                  |
| <input type="checkbox"/> <b>Foam pad</b>                          | 3 mm soft foam, about 30 × 45 mm, self-adhesive on one side.                                                         |
| <input type="checkbox"/> <b>Double-sided tape</b>                 | Thin; for the speaker and the face panel (clear transfer tape is best for the panel).                                |
| <input type="checkbox"/> <b>USB-C data cable</b>                  | One that carries data, not a charge-only cable.                                                                      |
| <input type="checkbox"/> <b>Blue paint pen (optional)</b>         | To colour the engraved wings and dart.                                                                               |

### Tools

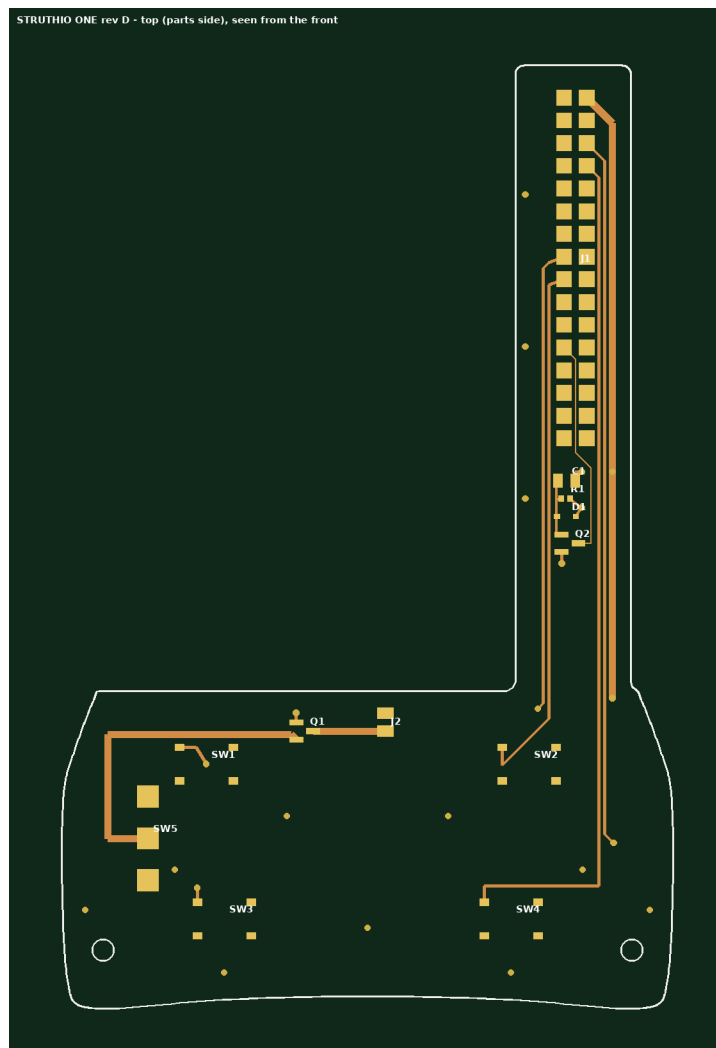
- ☐ A small Phillips screwdriver (PH0 or PH1) for the M2 screws
- ☐ Scissors, a craft knife
- ☐ A needle or pin (only if the battery plug needs its wires swapped)
- ☐ A paperclip (the side pin holes, for re-flashing later)
- ☐ A Windows 11 computer with internet access

# Order and make the parts

## PART 3

### The ONE board (package 3, JLCPCB)

1. On jlcpcb.com upload `struthio_one_gerbbers.zip` (rev D): 2 layers, 1.6 mm, any colour.
2. Turn on **PCB Assembly**, top side, **Standard** PCBA (three parts are through-hole). Upload `BOM_JLCPCB.csv` and `CPL_JLCPCB.csv`.
3. In the placement preview, compare with the picture below: **J1** an ordinary header on the strip; **J2** opening toward the right edge; **SW5** knob off the left edge; **Q1**, **Q2** single leg as on the silkscreen; **D1** band toward the left. Rotate any part that differs in 90° steps.



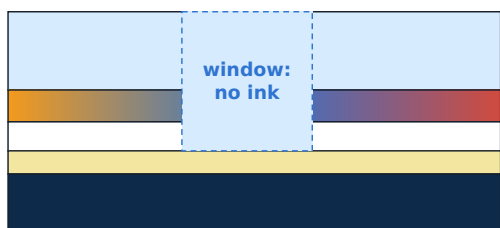
The board as ordered (top side, seen from the front of the handheld), with every part's place DIAGRAM

### The face panel (package 2)

Send the shop the files and this sentence: **1.0 mm clear cast acrylic, cut to the DXF, reverse-printed (colour, then white) from the mirrored PNG, window left clear.** Cheaper: a clear cut panel and the art printed on sticker vinyl, stuck on its back.



The proof: cut lines magenta, the clear window blue



clear acrylic 1.0 mm (the front you touch)

colour print (on its back)

white underprint

double-sided tape (black areas only)

front shell face

What the shop makes, in layers (the window stays clear) [DIAGRAM](#)

## The printed parts (package 4)

PETG (or ASA), 0.2 mm layers, 4 walls, 6 top and bottom layers, 40 % infill: solid walls. Not PLA (it softens in a hot car). One of each STL. The wing buttons and rocker print engraving-down: keep the first layer neat.

Side views, the bed at the bottom. No supports needed.



How each part sits on the bed [DIAGRAM](#)

# Flash the full game (Windows 11)

## PART 4 · DO THIS AS SOON AS THE WAVESHARE BOARD ARRIVES, BEFORE YOU BUILD IT IN

This puts the whole game on the board: the program, the art pack and the soundtrack. You need nothing but Python and one command. The board stays on its own on the desk, connected by USB-C.

### 4.1 Unzip the firmware package

Right-click `STRUTHIO_ONE_1_Firmware_Flashing.zip` → **Extract All...** → type `C:\` → Extract. You now have `C:\struthio`:

```
C:\struthio\  
├─ prebuilt\           ← the game, ready to flash (you run the command here)  
│   ├── bootloader.bin, partition-table.bin, struthio.bin  
│   └─ flash_args.txt  (addresses and settings for esptool)  
├─ handheld\build\assets\ ← the art pack and the soundtrack  
├─ manual\  
├─ STRUTHIO.bat        (the menu, for building it yourself)  
└─ README_FIRST.txt
```

Not inside OneDrive, and no spaces in the path: `C:\struthio` exactly.

### 4.2 Plug the board in

Connect the Waveshare to the PC with the USB-C **data** cable.

### 4.3 Double-click **FLASH\_ME.bat**

In `C:\struthio`, double-click **FLASH\_ME.bat**. The first time it installs Python and the flashing tool (allow it if Windows asks), then finds the board by itself, writes the game, its pictures and its music, and checks every byte.

"Windows protected your PC"? Click **More info**, then **Run anyway** (it is a plain text file you can read in Notepad).

✓ The window ends with DONE.

```
Terminal — ~/struthio/prebuilt (the emulator bench)

maker@bench:~/struthio/prebuilt$ python -m esptool --chip esp32s3 -p socket://localhost:5555 -b 460...
esptool v5.4.0

Connected to ESP32-S3 on socket://localhost:5555: 1
Chip type: ESP32-S3 (QFN56) (revision v0.3)

Stub flasher running.

Writing 'bootloader.bin' at 0x00000000...

Wrote 18704 bytes (12113 compressed) at 0x00000000 in 0.7 seconds (215.0 kbit/s).

Hash of data verified. 2

Writing 'struthio.bin' at 0x00010000...

Wrote 609936 bytes (395655 compressed) at 0x00010000 in 19.2 seconds (254.0 kbit/s).

Hash of data verified. 2

Writing '../handheld/build/assets/struthio.pak' at 0x00410000...

Wrote 8661084 bytes (1526098 compressed) at 0x00410000 in 84.8 seconds (817.1 kbit/s).

Hash of data verified. 2

Writing '../handheld/build/assets/struthio_music.ima' at 0x00d10000...

Wrote 1920020 bytes (1745564 compressed) at 0x00d10000 in 84.9 seconds (180.9 kbit/s).

Hash of data verified. 2

Hard resetting via RTS pin...
```

What FLASH\_ME runs underneath, by hand: the same esptool command (in `C:\struthio\prebuilt: py -m esptool --chip esp32s3 -p COM5 -b 460800 write-flash @flash_args.txt`). **1** connected; **2** each part written and checked. Captured against Espressif's ESP32-S3 emulator, so the port reads `socket://localhost:5555`; lines left out are marked ... [SCREENSHOT](#)



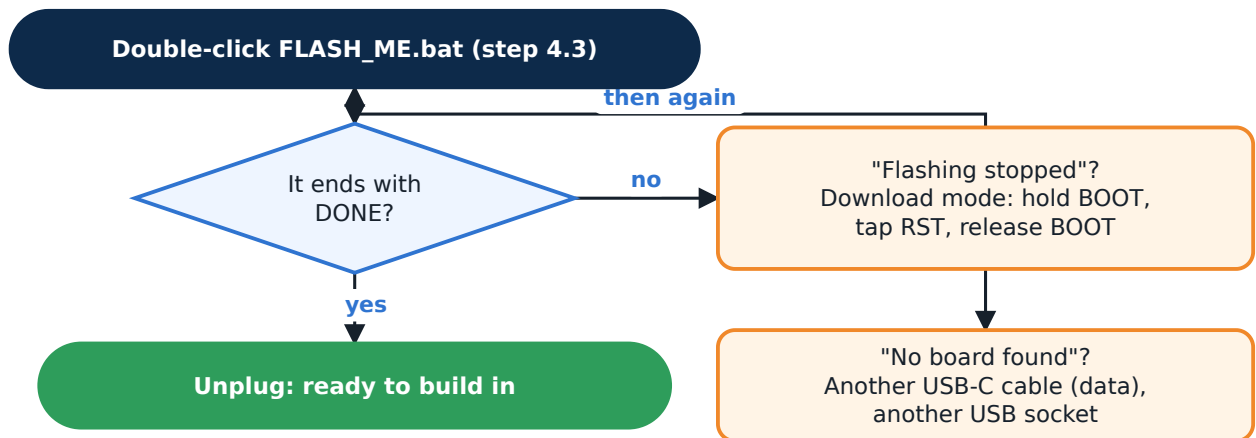
What the board shows: the game waiting at READY [SCREENSHOT](#)

## 4.4 Start the game

Unplug the board: it's ready to build in. (Pressed RST with no buttons fitted, it shows the first power-on check waiting for presses: that's correct.)

✓ FLASH\_ME said DONE.

## If flashing doesn't work



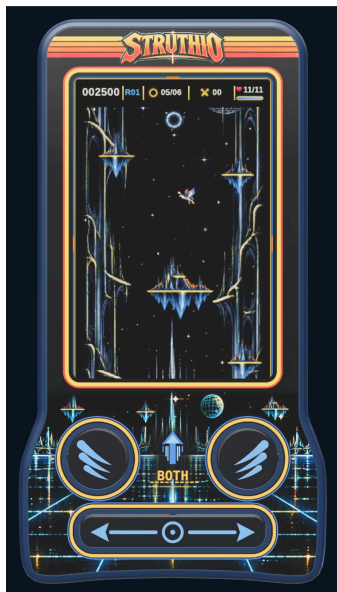
Flashing problems DIAGRAM

**Download mode** (only if it won't connect): hold the board's **BOOT** button, press and release **RST**, release **BOOT**, then run 4.6 again. FLASH\_ME looks for the board again by itself. After a later re-flash through the finished case, the same buttons are behind the pin holes on the left side (Part 8).

**Another way:** Espressif's browser flasher [espressif.github.io/esptool-js](https://espressif.github.io/esptool-js) (Chrome or Edge) does the same: add `prebuilt\bootloader.bin` at `0x0`, `partition-table.bin` at `0x8000`, `struthio.bin` at `0x10000`, `handheld\build\assets\struthio.pak` at `0x410000`, `struthio_music.ima` at `0xd10000`.

# Assemble

## PART 5 · STEPS 1-9 · WORK ON A SOFT CLOTH, NOTHING NEEDS FORCE



The engraved wings and dart, filled  
blue FROM THE 3D MODEL

### 1 Paint (optional)

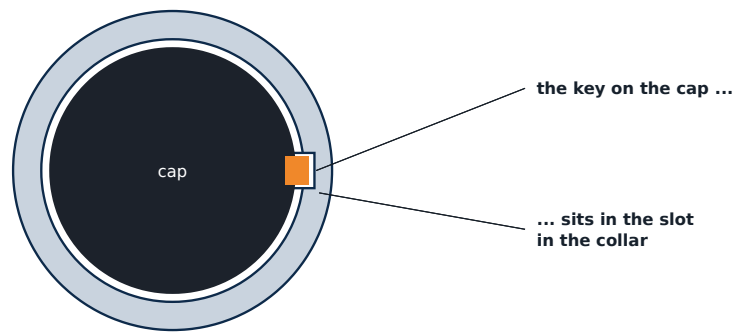
Fill the engraved wings and the dart with the blue paint pen. Wipe the top faces clean, let it dry.



### 2 Fit the wing buttons

From the **inside** of the front shell, push each round cap into its collar with its **key** (the small tab on its side) in the slot on the collar's outer side.

- ✓ Both caps are in, keys in their slots, wings upright from the front. Each one springs back when pressed.



Why it only goes in one way: a wing button seen from the inside of the front shell; the key points away from the middle

DIAGRAM



### 3 Fit the rocker

Push the rocker into its collar from the inside. Cut an **11 mm** piece of 1.75 mm filament and push it through the hole in the collar's top wall until it stops.

✓ The rocker tips freely both ways and springs back.



### 4 Lay in the Waveshare

Screen **down, USB-C at the top** (it lines up with the opening), its PWR/RST/BOOT buttons toward the pin holes on the left. It drops between the ribs.

✓ It lies flat and doesn't slide.

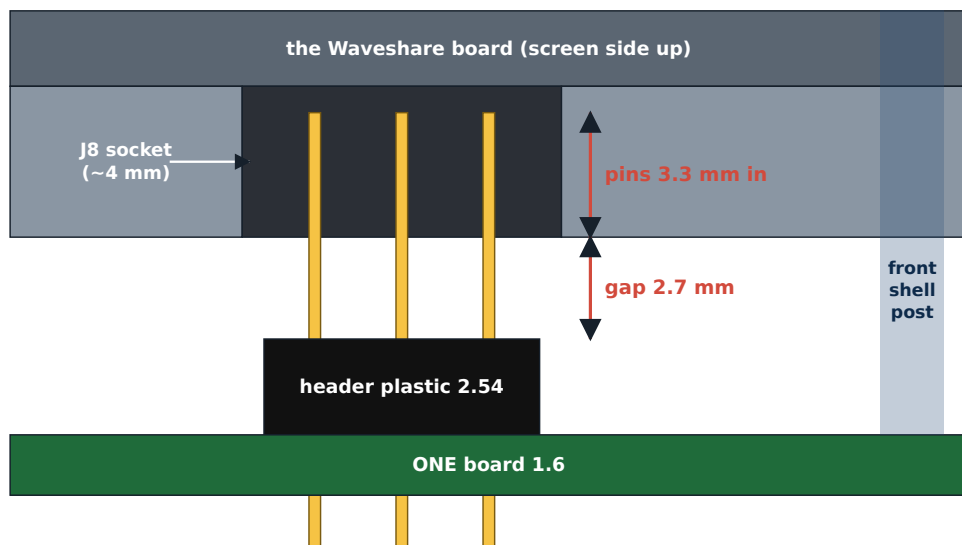


## 5 Plug on the ONE board

Set the slide switch **down (OFF)**. Hold the board over the Waveshare, header over the J8 socket, lower it straight and press evenly until it **rests on the four posts**.

It stops on the posts by itself. Don't press harder: the pins are meant to go only 3.3 mm in.

✓ The ONE board sits flat on all four posts.



How deep the header goes (side view, front at the top). The ONE board rests on the front shell's posts, and that sets the depth: the pins go 3.3 mm in and the plastic never touches the socket DIAGRAM

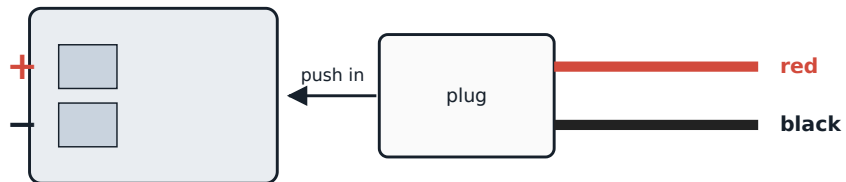


## 6 Battery

Check the plug against the + printed beside J2: **red to +**. Plug it into J2 from the right. Lay the cell on the Waveshare's back left of the strip, **lead end down**, the lead flat in the gap down to the plug.

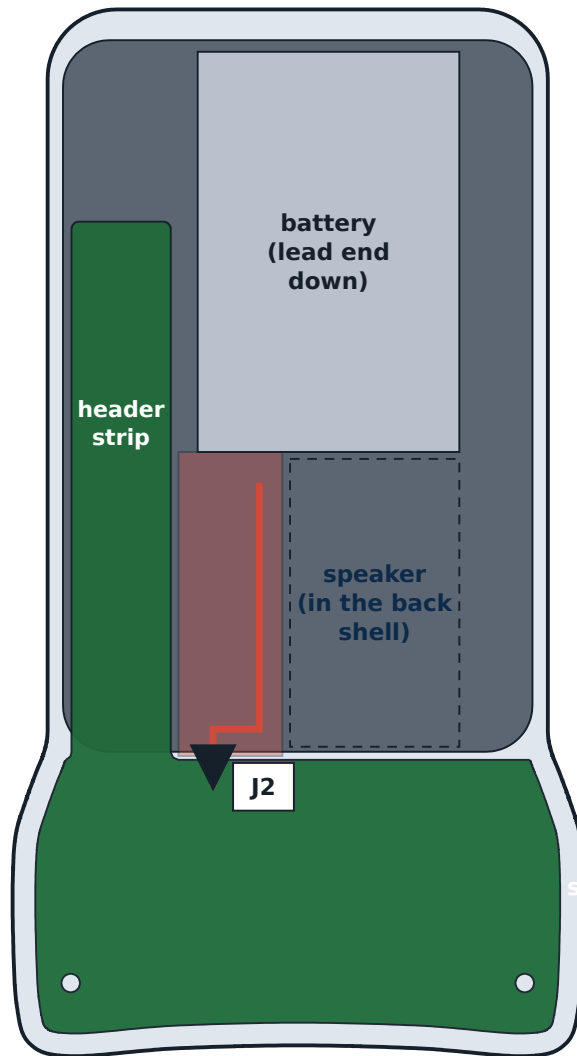
✓ Red wire on +, lead lying flat, nothing pinched.

J2 on the ONE board (white socket)

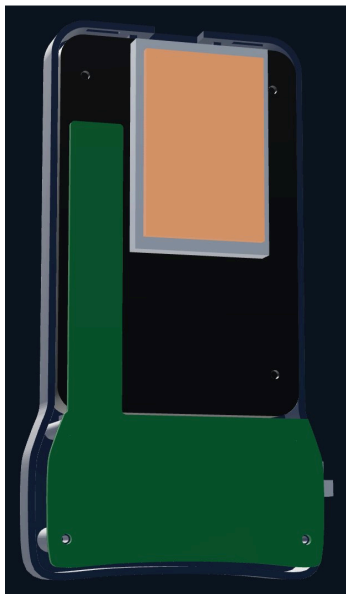


**RED wire on the + side**

Battery plug polarity. Plugged the wrong way round it does nothing (Q1 blocks it): swap the two crimp contacts (lift each latch with a needle) and plug in again [DIAGRAM](#)



Where things lie, seen from the back while you build (USB-C at the top). Red: the battery lead, flat on the Waveshare's back between the cell and the strip, down to J2 DIAGRAM



## 7 Foam pad

Stick the 3 mm foam pad on top of the battery. It presses the battery and the Waveshare forward when the case closes.



## 8 Speaker into the back shell

Tape on the speaker's back, stick it in the bay inside the back shell, **face toward the grille**. Plug its lead into **J9** on the Waveshare (small 2-pin socket at its left edge, beside BOOT). Keep the back shell right beside the case: the lead is short.

✓ Speaker in its bay, plug in J9.



## 9 Close the case

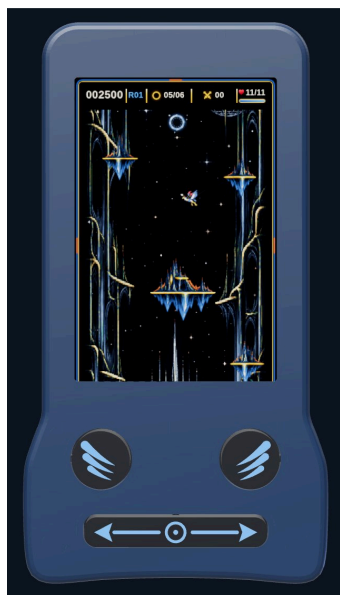
Put the back shell's two top hooks into the slots inside the front shell's top edge and swing it down. The power knob slides into its slot, three pegs drop into the Waveshare's holes. Fit the two M2 × 8 screws at the bottom, **snug, not tight**.

If it doesn't close by itself, open it and look: a lead or the foam is in the way. Never force it.

✓ Closed all round, no gap at the seam.

# First power and test

## PART 6 · STEPS 10-13

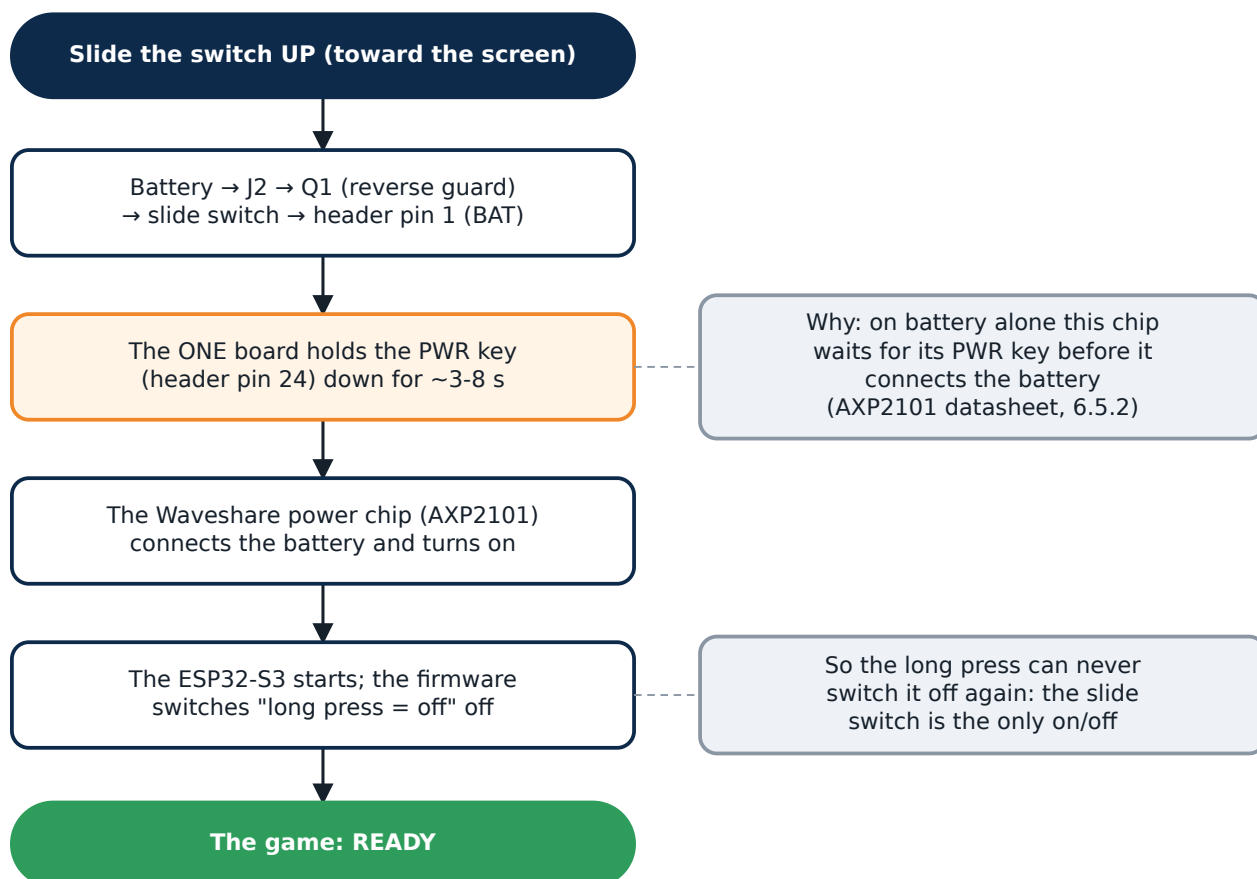


### 10 Switch on

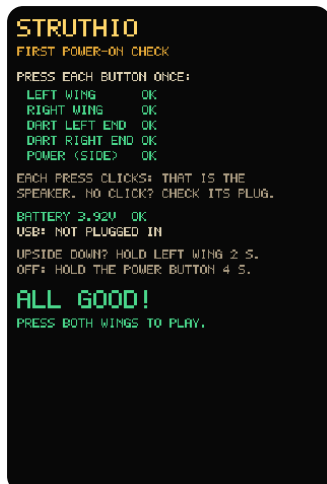
Slide the switch **up**. Within a few seconds the screen lights. The first time it shows the **first power-on check** (next step). Slide it down: off at once.

Left alone, it dims after 30 s and switches itself off after 5 minutes (slide off and on to wake it).

✓ The screen lights. (Nothing? Part 9.)



What happens when you slide the switch up **DIAGRAM**



The check, all done (example values, shown on a ONE SLIM: the ONE has no POWER row) [SCREENSHOT](#)

## 11 The first power-on check

Press each wing button and each rocker end **once**: each turns **OK** and clicks (the click is the speaker). **BATTERY** shows the cell. When all four are OK, press **both wings**: the game starts. The check never runs again.

STUCK DOWN next to a button: its cap rubs (Part 9). NO BATTERY: check the battery plug.

✓ ALL GOOD, then the game.

## 12 Picture the wrong way up?

On the check screen, **hold LEFT for 2 s**. Later, in service mode (both wings held while switching on), **hold RIGHT for 1 s**. Either is saved. The ONE board tells the firmware it is a ONE (header pin 9), so the 1000 mAh battery charges at 200 mA by itself.

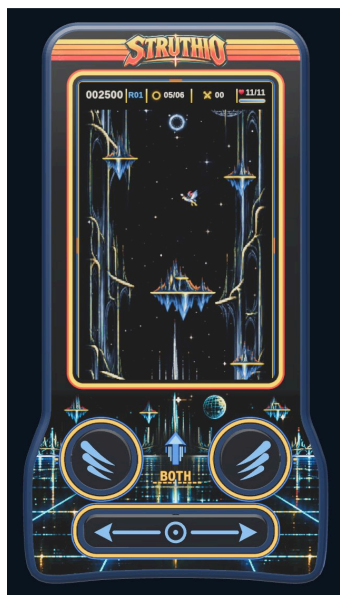
## 13 Charge

Plug USB-C in **with the switch ON**. With the switch OFF the battery is disconnected: USB still runs the game but doesn't charge.

✓ In service mode BATT says CHARGING and USB.

# Fit the face panel

## PART 7 · STEP 14, LAST



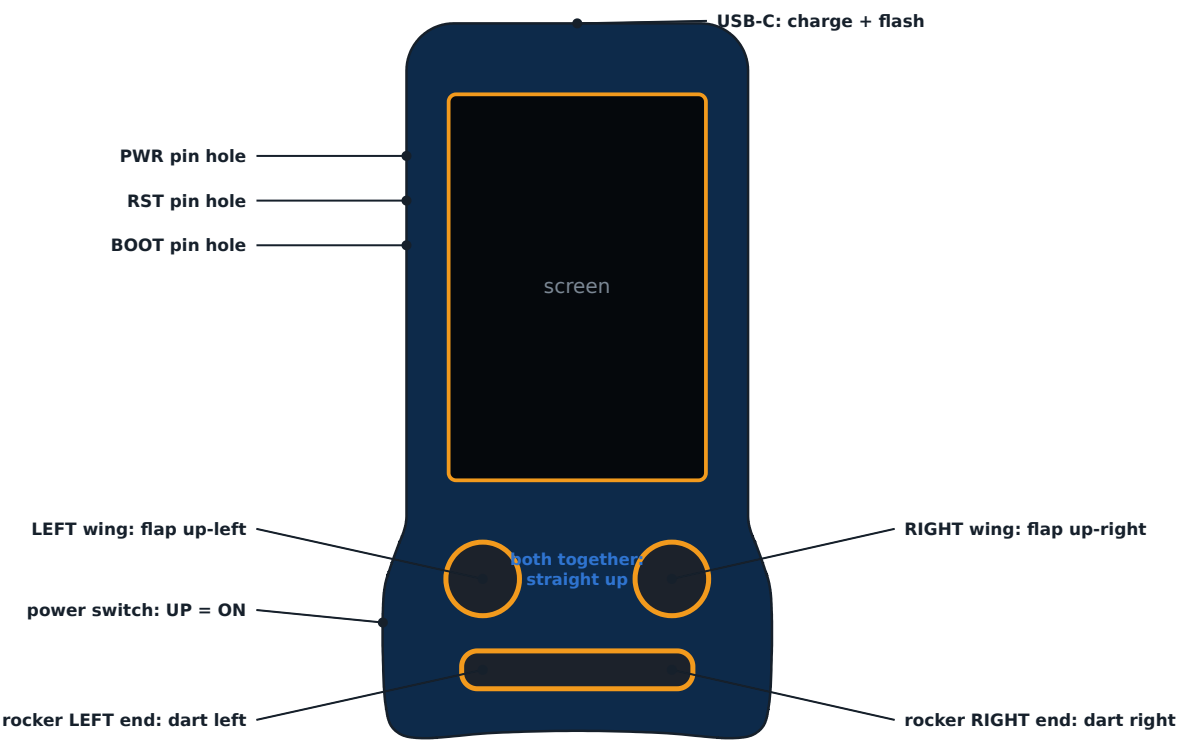
### 14 Face panel

Peel the protective film off the back. Tape on the **black areas only**. Line the panel up 2.4 mm in from the case's edge all round, the wells over the buttons, and press it down from the middle outward. Peel the front film.

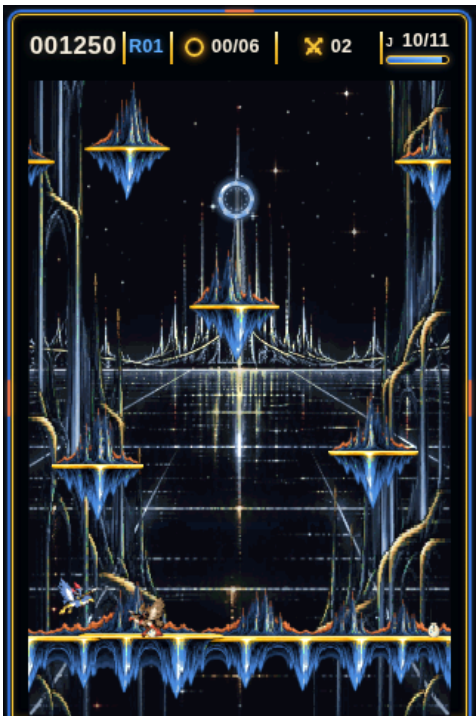
- ✓ The panel sits flat with an even border all round; every button moves freely in its well.

# How to play

## PART 8



The controls, and what is on the sides DIAGRAM



The game running SCREENSHOT

### The controls

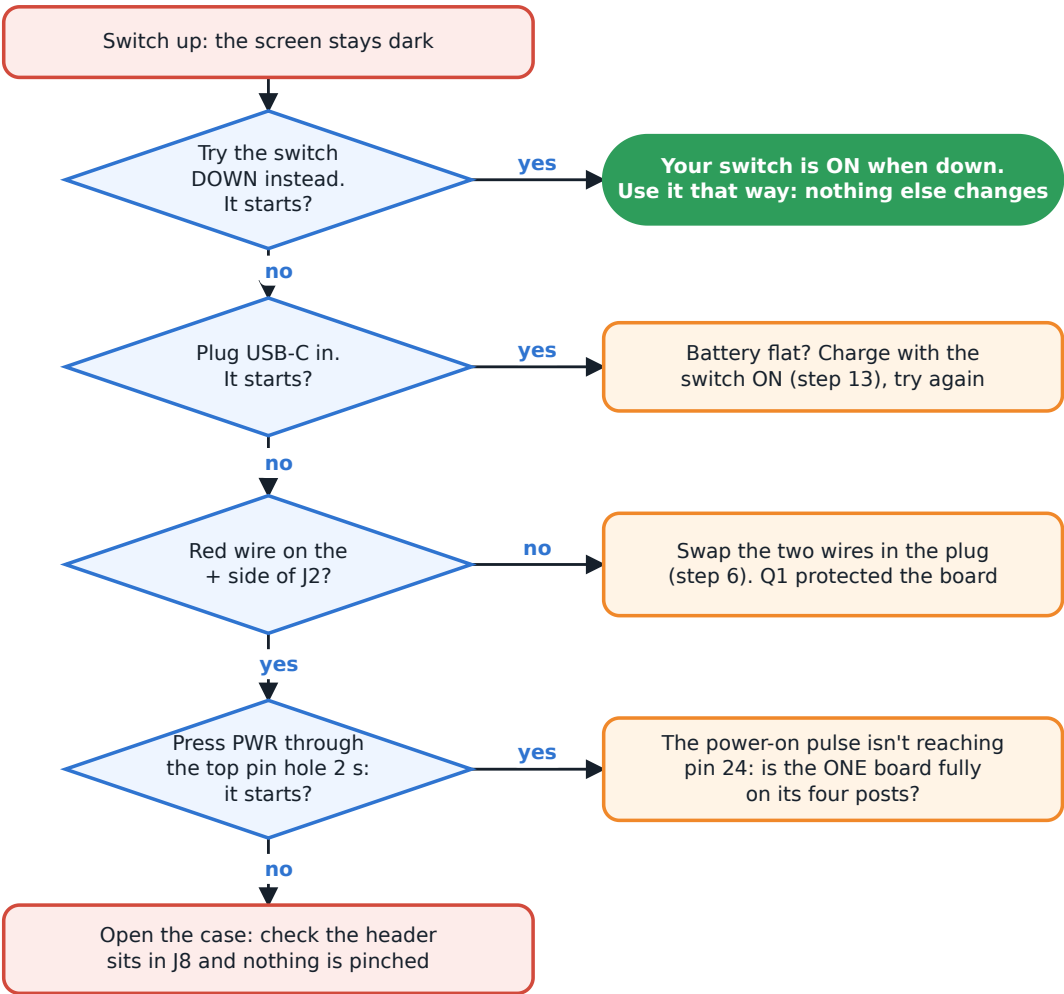
| Press                         | Does                        |
|-------------------------------|-----------------------------|
| LEFT wing                     | flap up and to the left     |
| RIGHT wing                    | flap up and to the right    |
| Both together                 | flap straight up            |
| Hold a wing                   | steer                       |
| Rocker left / right end       | aim-assisted dart, that way |
| Both held 1 s after GAME OVER | new run                     |
| Both held while switching on  | service mode                |

### Updating the game later

Same as Part 4 through the USB-C in the top edge; the switch can be on or off. Download mode through the case: a paperclip in the **BOOT** hole, tap **RST**, release.

# If something is wrong

## PART 9

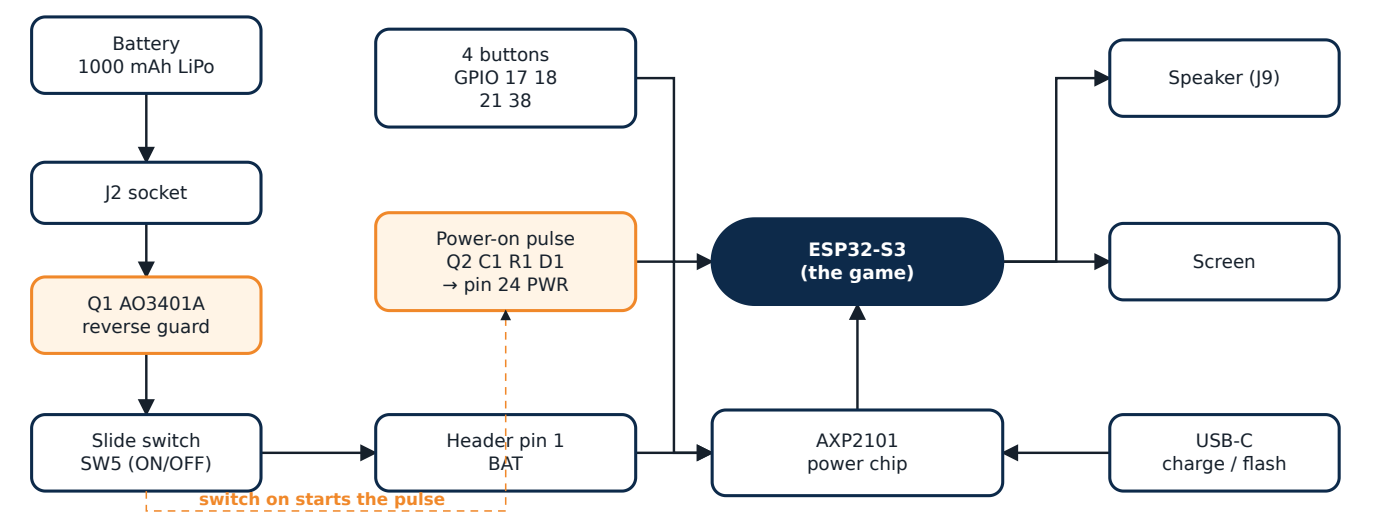


The screen stays dark when you switch on DIAGRAM

| What you see                                                 | What to do                                                                   |
|--------------------------------------------------------------|------------------------------------------------------------------------------|
| STUCK DOWN on the check screen, or a button counts by itself | Its stem touches the switch: take the cap out, sand 0.2 mm off the stem tip. |
| A button doesn't count                                       | Is the ONE board on all four posts? Does the cap move freely?                |
| No sound                                                     | Speaker plug in J9? Speaker face against the grille?                         |
| Back shell won't close                                       | A lead or the foam is in the way. Knob in its slot? Never force it.          |
| Picture upside down                                          | Service mode, hold RIGHT 1 s.                                                |
| Flashing fails                                               | Part 4, "If flashing doesn't work".                                          |

# How it works

## PART 10



The electrics: what the ONE board does DIAGRAM



The 23 mm stack, front to back DIAGRAM

## The ONE board

| Part      | Job                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------|
| J1 header | into the Waveshare J8: BAT, GND, 4 buttons, PWR, and pin 9 to GND (tells the firmware it is a ONE) |
| SW1-SW4   | the wing buttons and rocker ends                                                                   |
| SW5       | the slide switch: a real battery cut                                                               |
| J2        | the battery plug                                                                                   |

|                   |                                |
|-------------------|--------------------------------|
| Q1                | reverse-battery guard          |
| Q2, C1, R1,<br>D1 | power-on pulse on PWR (pin 24) |

## Header pins used

1 BAT · 3, 4, 29, 30 GND · 9 GND (model strap) · 5 GPIO21 (rocker L) · 7 GPIO38 (rocker R) · 16 GPIO17 (left wing) · 18 GPIO18 (right wing) · 24 PWR

## Things to confirm on the first build

| What                | Designed for                       | If different                                                    |
|---------------------|------------------------------------|-----------------------------------------------------------------|
| J8 socket depth     | ~4 mm, pins 3.3 mm in              | pins must hold; a 0.3 mm shim under the posts changes the depth |
| Waveshare hole size | ≥ 2.2 mm (pegs 1.8)                | sand the pegs                                                   |
| Speaker size, lead  | bay 22 × 37 × 7.5 mm, lead ≥ 60 mm | any box that fits, with tape                                    |
| Switch ON direction | up                                 | use it the other way                                            |
| USB-C height        | opening 12.8 × 7.8 mm              | file it wider                                                   |
| Acrylic thickness   | 1.0 ± 0.1 mm                       | the panel may sit a hair proud                                  |

STRUTHIO ONE build manual · commit ba90005 · 2026-10-02. Made from the design files by [tools/manual/one\\_manual.py](#).